

A quadratic-partition counterexample to the Pierce–Birkhoff conjecture

Abstract

We construct a continuous piecewise quadratic function on all of $\mathbb{R}^8$ whose pieces are determined by finitely many polynomial sign conditions of total degree at most two, and which admits no finite maximum–minimum expression in polynomials of arbitrary degree. The construction adds two independent cofactor coordinates to a six-variable example. A quantitative determinant estimate gives a continuous extension on a quadratic partition, and polynomial restriction proves nonrepresentation. Consequently the hyperplane partition theorem is sharp with respect to partition degree.

1 Statement

There is a continuous function on all of $\mathbb{R}^8$, with one polynomial label on every entire realization of the signs of a finite family of quadratic polynomials, that has no finite polynomial maximum–minimum representation. Its selected polynomial labels themselves have degree at most two. All eight coordinates are independent.

The construction below gives explicit partition data and a global continuity proof. Nonrepresentation follows by an exact polynomial pullback to the six-dimensional function, whose arbitrary-degree nonrepresentation theorem is proved in Appendix A. The required ambient extension is proved here; no general extension from polynomial graphs is assumed.

2 The six-dimensional polynomial data

Write $w = (x, y, z, u, v, r) \in \mathbb{R}^6$, and put

$$C(w) = \begin{pmatrix} x & -v + r/2 & -y \\ y & z + u & x \\ z & r/2 & 0 \\ u & -v/2 + r & -v/2 \\ v & 2z + u/2 & u/2 \end{pmatrix}, \quad J = \text{diag}(1, 1, 3, -1, -1). \quad (\text{Q1})$$

Let A be the first three rows of C , let B be the last two rows, and define

$$Q = C^T J C = A^T \text{diag}(1, 1, 3) A - B^T B, \quad D = \det A. \quad (\text{Q2})$$

Direct multiplication has the form

$$Q = \begin{pmatrix} a & b & 0 \\ b & d - a & e \\ 0 & e & d \end{pmatrix}. \quad (\text{Q3})$$

Here a, b, d, e denote the displayed entries of Q , not additional independent variables. Define four homogeneous quadratic polynomials

$$\begin{aligned}
q_1 &= -10b + 2d + 2e, \\
q_2 &= 4a + 2b - 2d - 2e, \\
q_3 &= -4a + 2b + 6d - 10e, \\
q_4 &= -4a + 2b + 2d + 2e,
\end{aligned}
\quad h = \min(q_1, q_2, q_3, q_4). \tag{Q4}$$

The six-dimensional function used below is

$$f_6(w) = \begin{cases} h(w), & h(w) > 0 \text{ and } D(w) > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{Q5}$$

The nonrepresentation theorem in Appendix A allows arbitrary real coefficients and arbitrary finite degrees of the representing polynomials.

For completeness, the quantitative Gram bound required below follows from an explicit constant matrix identity. Set

$$\begin{aligned}
s_i &= 4e_i - \mathbf{1} \quad (0 \leq i < 3), & s_3 &= -\mathbf{1}, \\
t_i &= (e_i + \mathbf{1})/4 \quad (0 \leq i < 3), & t_3 &= 0.
\end{aligned}$$

For $q_{ij} = -s_i^T Q s_j$, the four retained edges 01, 03, 12, 13 are respectively q_1, q_2, q_3, q_4 , and

$$q_{02} = 2q_1 + 6q_2 + 5q_4, \quad q_{23} = q_1 + 2q_2 + 2q_4. \tag{Q6}$$

The exact reconstruction and constant identities are

$$Q = \sum_{i < j} q_{ij} (t_i - t_j)(t_i - t_j)^T, \quad \sum_{i < j} (t_i - t_j)(t_i - t_j)^T = \frac{I + \mathbf{1}\mathbf{1}^T}{4}. \tag{Q7}$$

When $h > 0$, every coefficient q_{ij} is at least h . Consequently

$$Q \succeq \frac{h}{4} I. \tag{Q8}$$

All entries and identities (Q1)–(Q8) are over the original six coordinates.

3 A quantitative determinant bound

Write the third row of A as

$$a_3 = (z, r/2, 0).$$

Whenever $h > 0$,

$$|D| \geq \frac{h}{4} \sqrt{z^2 + r^2/4}. \tag{Q9}$$

To prove this, put

$$M = \text{diag}(1, 1, \sqrt{3})A, \quad \alpha = h/4 > 0.$$

Equations (Q2) and (Q8) give $M^T M = Q + B^T B \succeq \alpha I$. Let $0 < \sigma_1 \leq \sigma_2 \leq \sigma_3$ be the singular values of M . Then $\sigma_1, \sigma_2 \geq \sqrt{\alpha}$, and hence

$$|\det M| = \sigma_1 \sigma_2 \sigma_3 \geq \alpha \sigma_3 \geq \alpha \|\text{third row of } M\| = \alpha \sqrt{3} \|a_3\|.$$

Since $|\det M| = \sqrt{3}|D|$, division by $\sqrt{3}$ proves (Q9). Moreover A is invertible, so $a_3 \neq 0$.

4 Independent auxiliary coordinates and the function on all of eight-dimensional space

Adjoin two independent real coordinates s, t . The two relevant cofactors of A are the homogeneous quadratic polynomials

$$c_1 = x(-v + r/2) + y(z + u), \quad c_2 = -(x^2 + y^2). \quad (\text{Q10})$$

Expansion along the third row gives the exact identity

$$D = zc_1 + (r/2)c_2. \quad (\text{Q11})$$

Define the quadratic polynomials

$$E_1 = s - c_1(w), \quad E_2 = t - c_2(w), \quad \tilde{D} = zs + (r/2)t. \quad (\text{Q12})$$

For $1 \leq j \leq 4$, $1 \leq k \leq 2$, and $\epsilon \in \{-1, 1\}$, put

$$L_{j,k,\epsilon} = q_j + 8\epsilon E_k. \quad (\text{Q13})$$

There are sixteen such polynomial labels. They all have total degree at most two in the eight independent coordinates. Put

$$H = \min_{j,k,\epsilon} L_{j,k,\epsilon} = h - 8 \max(|E_1|, |E_2|). \quad (\text{Q14})$$

Define on every point of $\mathbb{R}^8$

$$F(w, s, t) = \begin{cases} H(w, s, t), & H(w, s, t) > 0 \text{ and } \tilde{D}(w, s, t) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{Q15})$$

No equation $E_k = 0$ is imposed on the domain of (Q15).

5 The quadratic orientation cannot vanish where H is positive

Suppose $H > 0$. Equation (Q14) implies $h > 0$ and

$$|E_1| < h/8, \quad |E_2| < h/8. \quad (\text{Q16})$$

By (Q11)–(Q12),

$$\tilde{D} - D = zE_1 + (r/2)E_2.$$

Cauchy–Schwarz, (Q16), and $\sqrt{2} < 2$ give

$$\begin{aligned} |\tilde{D} - D| &\leq \sqrt{z^2 + r^2/4} \sqrt{E_1^2 + E_2^2} \\ &< \sqrt{z^2 + r^2/4} \frac{\sqrt{2}h}{8} \\ &< \sqrt{z^2 + r^2/4} \frac{h}{4} \leq |D|. \end{aligned} \quad (\text{Q17})$$

The third-row norm is positive here by Section 3. Thus

$$H > 0 \implies \tilde{D} \neq 0 \text{ and } \text{sgn } \tilde{D} = \text{sgn } D. \quad (\text{Q18})$$

This estimate holds globally in all eight independent coordinates.

6 Global continuity and an explicit finite quadratic partition

On the open set $\{H > 0\}$, equation (Q18) says that $\tilde{D}$ never vanishes. Its sign is therefore locally constant, and (Q15) is locally either the continuous function H or zero. On $\{H < 0\}$ it is identically zero. At every point where $H = 0$, continuity follows from

$$0 \leq F \leq \max(H, 0). \quad (\text{Q19})$$

This proves continuity on all of $\mathbb{R}^8$.

Here are explicit finite partition polynomials. Enumerate the sixteen labels in (Q13) in any fixed order as $L_1, \dots, L_{16}$. Use

$$\tilde{D}, \quad L_a \quad (1 \leq a \leq 16), \quad L_a - L_b \quad (1 \leq a < b \leq 16). \quad (\text{Q20})$$

This list has at most $1 + 16 + 120 = 137$ polynomials, each of total degree at most two. Omit identically zero or constant polynomials and repetitions; their known signs require no partition test.

For each nonempty *entire* realization of their signs, assign a label as follows. If its sign of $\tilde{D}$ is positive and every sign of L_a is positive, choose the smallest index among those attaining $\min_a L_a$. Otherwise assign the zero polynomial. The comparisons in (Q20) determine the same selected index at every point of the entire sign realization, including all of its disconnected components. Thus this rule gives one polynomial on the whole realization. Zero-sign realizations cause no ambiguity: where some $L_a \leq 0$, (Q15) is zero; where $\tilde{D} = 0$, equation (Q18) excludes $H > 0$, so (Q15) is again zero. Ties among the minimizing labels are resolved by the fixed index order and do not change their common numerical value.

Equivalently, a closed cover uses

$$T_0 = \{\tilde{D} \leq 0\} \cup \bigcup_a \{L_a \leq 0\}$$

with label zero, and

$$T_a = \{\tilde{D} \geq 0, L_b \geq 0 \forall b, L_a \leq L_b \forall b\} \quad (\text{Q21})$$

with label L_a . These sets cover the whole space and their labels agree on overlaps. At $\tilde{D} = 0$ with all $L_b \geq 0$, (Q18) forces their minimum to be zero.

7 Exact pullback and arbitrary-degree nonrepresentation

Define the polynomial map

$$\gamma : \mathbb{R}^6 \longrightarrow \mathbb{R}^8, \quad \gamma(w) = (w, c_1(w), c_2(w)). \quad (\text{Q22})$$

On this graph, $E_1 = E_2 = 0$. Therefore (Q11)–(Q15) give the exact identity on all of $\mathbb{R}^6$

$$F \circ \gamma = f_6. \quad (\text{Q23})$$

Suppose for contradiction that

$$F = \max_{1 \leq i \leq N} \min_{1 \leq j \leq M_i} P_{ij}$$

for finitely many polynomials $P_{ij} \in \mathbb{R}[x, y, z, u, v, r, s, t]$, with unrestricted real coefficients and degrees. Composition with the polynomial map (Q22) commutes pointwise with finite maximum and minimum, so (Q23) yields

$$f_6 = \max_{1 \leq i \leq N} \min_{1 \leq j \leq M_i} (P_{ij} \circ \gamma).$$

Every $P_{ij} \circ \gamma$ is a polynomial in the original six independent coordinates. This contradicts the arbitrary-degree nonrepresentation of (Q5), proved in Appendix A. Consequently no such finite polynomial expression for F exists.

The result is a counterexample to the unrestricted quadratic-partition statement. It asserts an example in dimension eight, not a minimal dimension. Since the hyperplane-partition theorem allows unrestricted piece degrees, its partition-degree bound is sharp [1]. Thus degree two is the smallest partition degree for which a counterexample exists.

References

- [1] Z. Lai and L.-H. Lim, *Pierce–Birkhoff conjecture is true for splines*, arXiv:2507.07976, 2025. <https://arxiv.org/abs/2507.07976>.

A The complete six-variable nonrepresentation argument

In this appendix f denotes the six-variable function called f_6 in the main text. The coordinates, matrix, and four labels are the same. The proof below establishes nonrepresentation for arbitrary finite families of real polynomials, with no degree bound.

A.1 Statement and the whole-space function

The construction in this note is a continuous piecewise quadratic function on all of $\mathbb{R}^6$, with five polynomial labels including zero. It has no finite expression in arbitrary real polynomials using pointwise minimum and maximum. The number, coefficients and degrees of those hypothetical polynomials are unrestricted.

Use coordinates $w = (x, y, z, u, v, r)$ on $W = \mathbb{R}^6$. On $U = \mathbb{R}^5$, use the bilinear form

$$\beta = \text{diag}(1, 1, 3, -1, -1),$$

which has signature $(3, 2)$. Define the following matrix of three columns:

$$C(w) = (c_0, c_1, c_2) = \begin{pmatrix} x & -v + r/2 & -y \\ y & z + u & x \\ z & r/2 & 0 \\ u & -v/2 + r & -v/2 \\ v & 2z + u/2 & u/2 \end{pmatrix}. \quad (1)$$

Let A be its first three rows, let B be its last two rows, and put

$$Q = C^T \beta C = A^T \text{diag}(1, 1, 3)A - B^T B, \quad D = \det A. \quad (2)$$

These definitions use all six independent coordinates. In particular, c_0 is the projection onto the first five coordinates, and the map $w \mapsto C(w)$ is injective.

Direct multiplication gives

$$Q = \begin{pmatrix} a & b & 0 \\ b & d - a & e \\ 0 & e & d \end{pmatrix}, \quad a = Q_{00}, \quad b = Q_{01}, \quad d = Q_{22}, \quad e = Q_{12}. \quad (3)$$

Thus the two identities $Q_{00} + Q_{11} - Q_{22} = 0$ and $Q_{02} = 0$ hold on the whole space. Define four homogeneous quadratic polynomials by

$$\begin{aligned} q_1 &= -10b + 2d + 2e, \\ q_2 &= 4a + 2b - 2d - 2e, \\ q_3 &= -4a + 2b + 6d - 10e, \\ q_4 &= -4a + 2b + 2d + 2e. \end{aligned} \quad (4)$$

Write $h = \min(q_1, q_2, q_3, q_4)$. The fixed function is

$$f(w) = \begin{cases} h(w), & h(w) > 0 \text{ and } D(w) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The positive diagonal coefficient 3 in (2) causes no restriction on the underlying real quadratic space: multiplying the third row of A by $\sqrt{3}$ gives the usual form $A^T A - B^T B$, with the same determinant sign. All coefficients in (1), (3), and (4) are rational.

A.2 Positive Gram margin, continuity, and a finite closed cover

Here is a matrix identity establishing that the strict inequalities in (4) imply $Q \succ 0$. Let $\mathbf{1} = (1, 1, 1)^T$, and put

$$a_i = 4e_i - \mathbf{1} \quad (0 \leq i < 3), \quad a_3 = -\mathbf{1},$$

$$u_i = (e_i + \mathbf{1})/4 \quad (0 \leq i < 3), \quad u_3 = 0.$$

For all six pairs $i < j$, set $q_{ij} = -a_i^T Q a_j$. The identities $\sum a_i = 0$ and $\sum u_i a_i^T = I_3$ give

$$Q = \sum_{i < j} q_{ij} (u_i - u_j)(u_i - u_j)^T. \quad (6)$$

For example, this also follows by multiplying out: the matrix $(a_i^T Q a_j)_{i,j=0}^3$ has zero row sums and off-diagonal entries $-q_{ij}$, so its quadratic expression is the sum of the corresponding six squared differences.

The four polynomials in (4) are respectively $q_{01}, q_{03}, q_{12}, q_{13}$. On the matrix space (3), the other two satisfy

$$q_{02} = 2q_{01} + 6q_{03} + 5q_{13}, \quad q_{23} = q_{01} + 2q_{03} + 2q_{13}. \quad (7)$$

Consequently $h > 0$ makes all six coefficients in (6) strictly positive. The three vectors $u_i - u_3$, $i < 3$, span $\mathbb{R}^3$, so (6) implies $Q \succ 0$. If $A\xi = 0$ for a nonzero ξ , (2) instead gives $\xi^T Q \xi = -|B\xi|^2 \leq 0$. We have proved

$$h(w) > 0 \implies Q(w) \succ 0 \implies D(w) \neq 0. \quad (8)$$

The function (5) is therefore continuous on $\{h > 0\}$, since the determinant sign is locally constant there. It is zero on $\{h < 0\}$, and continuity at $h = 0$ follows from $0 \leq f \leq \max(h, 0)$.

An explicit closed semialgebraic cover of the whole space consists of

$$F_0 = \{D \leq 0\} \cup \bigcup_{i=1}^4 \{q_i \leq 0\},$$

with polynomial label zero, and

$$F_i = \{D \geq 0, q_j \geq 0 \ (1 \leq j \leq 4), q_i \leq q_j \ (1 \leq j \leq 4)\}, \quad 1 \leq i \leq 4, \quad (9)$$

with label q_i . If $D = 0$ and all four labels are nonnegative, (8) forces their minimum to be zero. This verifies the labels on every boundary overlap. The five sets in (9), including F_0 , cover all of W . Furthermore

$$f(\lambda w) = \lambda^2 f(w) \quad (\lambda > 0). \quad (10)$$

For an additional direct check that the Gram-positive set is nonempty, at $w = (2, 8, 0, -1, -7, -9)$ the Gram matrix is

$$\begin{pmatrix} 18 & -12 & 0 \\ -12 & 75/2 & -3 \\ 0 & -3 & 111/2 \end{pmatrix},$$

whose leading principal minors are 18, 531, 58617/2.

A.3 The polynomial Gram-zero curve and all its quadratic equations

Define the polynomial curve

$$b(t) = \begin{pmatrix} t^4 - 6t^2 + 1 \\ 4t^3 - 4t \\ t^4 + 2t^2 + 1 \\ 2t^4 - 2 \\ 4t^3 + 4t \\ 2t^5 + 4t^3 + 2t \end{pmatrix}, \quad k(t) = (-t, 1, 1)^T, \quad R_- = (0, 1, -1)^T(0, 1, -1). \quad (11)$$

Write $d_*(t) = 12(1 + t^2)^2$. The following are exact polynomial identities:

$$Q(b(t)) = 0, \quad C(b(t))k(t) = 0, \quad Q(b'(t)) = d_*(t)R_-. \quad (12)$$

In particular, the column relation is $c_1(b(t)) = tc_0(b(t)) - c_2(b(t))$. Its coefficient of c_2 is constantly -1 ; this fact permits the rank-one target R_- throughout the curve.

The coefficient matrix of the six entries of $b(t)$, with rows indexed by degrees zero through five, is

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 & 0 \\ 0 & -4 & 0 & 0 & 4 & 2 \\ -6 & 0 & 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}. \quad (13)$$

Its determinant is $2048 = 2^{11}$. Thus no nonzero real homogeneous linear polynomial vanishes identically on the curve. The curve, viewed at a transcendental ordered parameter, also has no such linear equation.

Let I_2 be the real vector space of homogeneous quadratics q in the six original variables satisfying $q(b(t)) \equiv 0$. Let

$$H_2 = \{q \in I_2 : q(b'(t)) \equiv 0\}.$$

The exact spaces needed below are

$$\dim I_2 = 10, \quad H_2 = \text{span}_{\mathbb{R}}\{Q_{00}, Q_{01}, Q_{12} + Q_{22}\}. \quad (14)$$

These statements concern every real homogeneous quadratic, including all the additional quadratics outside the Gram span. Here is a finite integer certificate of (14). Order the 21 monomials $w_i w_j$ lexicographically by $0 \leq i \leq j < 6$. Form the 11×21 matrix M whose column (i, j) contains the coefficients, in degrees zero through ten, of $b_i(t)b_j(t)$. Form the 9×21 matrix M' from the coefficients, in degrees zero through eight, of $b'_i(t)b'_j(t)$, and put $N = \begin{pmatrix} M \\ M' \end{pmatrix}$. With all indices beginning at zero:

- The minor of M using all rows and columns $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 20$ has determinant 2^{35} .
- The minor of N using rows $0, 1, \dots, 17$ and columns $0, 1, \dots, 16, 20$ has determinant 2^{69} .

The first assertion also follows directly from (13): the coordinate polynomials span all polynomials of degree at most five, so their products span all polynomials of degree at most ten. Thus $\text{rank } M = 11$ and $\dim I_2 = 21 - 11 = 10$.

By (12), all three displayed Gram polynomials in (14) lie in $\ker N$. They are independent: their coefficients on monomials x^2, xz, xv form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix},$$

with determinant one. Hence $\text{rank } N \leq 18$, while the second minor gives the reverse inequality. This proves (14) without assuming that the four Gram entries generate the full space of quadratic equations.

A.4 First and second variations inside the fixed six-dimensional space

For a homogeneous quadratic matrix map, write $dQ_b(z)$ for its linear derivative at b in direction z . In particular,

$$Q(b + \varepsilon z) = Q(b) + \varepsilon dQ_b(z) + \varepsilon^2 Q(z) \quad (15)$$

is an exact identity.

Fix arbitrary real t and $\rho > 0$, and take the base $b_* = \rho b(t)$. In this section all derivatives of the curve are evaluated at that fixed t . Set

$$n = -\frac{b''}{2\rho d_*}, \quad j_0 = b''' - \frac{6t}{1+t^2} b''. \quad (16)$$

Differentiating (12) twice gives $dQ_{b(t)}(b'') = -2d_* R_-$. Differentiating a third time gives $dQ_{b(t)}(b''') = -3d'_* R_-$. Since $3d'_*/(2d_*) = 6t/(1+t^2)$, it follows that

$$dQ_{b_*}(n) = R_-, \quad dQ_{b_*}(j_0) = 0. \quad (17)$$

For a symmetric matrix in the four-dimensional space (3), define

$$\pi(R) = (R_{00}, R_{01}, R_{22}), \quad \ell(R) = k^T R k. \quad (18)$$

The maps π and ℓ together identify that matrix space with $\mathbb{R}^4$: explicitly,

$$\ell(R) = (t^2 - 1)R_{00} - 2tR_{01} + 2R_{22} + 2R_{12}. \quad (19)$$

The equation $C(b_*)k = 0$ implies

$$\ell(dQ_{b_*}(z)) = 0 \quad (z \in W). \quad (20)$$

Let e_x, e_y, e_z be the first three standard coordinate vectors of W , and let J_3 have columns $\pi(dQ_{b_*}(e_x)), \pi(dQ_{b_*}(e_y)), \pi(dQ_{b_*}(e_z))$. An exact derivative calculation gives

$$\det J_3 = 12\rho^3(1+t^2)^6 > 0. \quad (21)$$

In particular dQ_{b_*} has rank exactly three, its image is precisely $\ker \ell$, and its three displayed columns give coordinates on that image. No generic rank hypothesis is being imposed on the chosen base.

The following identities supply the determinant direction and the second variation missing from (20):

$$\begin{aligned} \beta(C(n)k, C(n)k) &= 0, \\ \beta(C(n)k, C(j_0)k) &= 0, \\ \beta(C(j_0)k, C(j_0)k) &= 1728, \\ dD_{b_*}(n) &= 0, \\ dD_{b_*}(j_0) &= -48\rho^2(1+t^2)^4. \end{aligned} \quad (22)$$

For the first two identities there is also a direct geometric explanation. Write $v_0(t) = c_0(b(t))$. For $K = \text{diag}(J, 0, J/2)$, with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, one has

$$v'_0 = \frac{4tv_0 - 4Kv_0}{1 + t^2}, \quad C(b'')k = 2v'_0. \quad (23)$$

Thus $C(n)k$ lies in the isotropic plane spanned by v_0, Kv_0 . These two vectors are independent: the third coordinate of v_0 is $(1 + t^2)^2 > 0$, the third coordinate of Kv_0 is zero, and $x(b(t))^2 + y(b(t))^2 = (1 + t^2)^4 > 0$ makes $Kv_0 \neq 0$. By (17), $C(j_0)k$ is perpendicular to that plane. Its positive squared norm and the two determinant derivatives in (22) are the exact values given by (1) and (11).

Fix once and for all the positive matrix

$$R_0 = \begin{pmatrix} 1 & 1/4 & 0 \\ 1/4 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \quad (24)$$

It satisfies (3), and

$$s = \ell(R_0) = t^2 - t/2 + 3 = (t - 1/4)^2 + 47/16 > 0.$$

Normalize the determinant direction by

$$j = -\sqrt{s/1728} j_0. \quad (25)$$

Equations (17) and (22) now give

$$\begin{aligned} dQ_{b_*}(j) &= 0, & dD_{b_*}(j) &> 0, \\ \ell(Q(n \pm \delta j)) &= \delta^2 s & \text{for every real } \delta. \end{aligned} \quad (26)$$

In particular, the two signs in (26) have exactly the same required second variation, even when the coefficient δ is chosen arbitrarily small.

A.5 Corrected paths with exactly the same singularly scaled Gram target

Fix t, ρ as above and then fix any $\delta > 0$. Put $y_{\pm} = n \pm \delta j$. The four-dimensional space

$$E = \text{span}(e_x, e_y, e_z, j)$$

has the displayed basis: its first three vectors map independently under dQ_{b_*} , while j maps to zero and has a nonzero determinant derivative.

For $z \in E$, consider the following four real equations:

$$\Phi_{\pm}(\varepsilon, z) = \begin{pmatrix} \pi(dQ_{b_*}(z) + \varepsilon[Q(y_{\pm} + z) - \delta^2 R_0]) \\ \ell(Q(y_{\pm} + z) - \delta^2 R_0) \end{pmatrix} = 0. \quad (27)$$

At $(\varepsilon, z) = (0, 0)$, these equations hold by (26). In the displayed basis of E , their derivative with respect to z has the block form

$$\begin{pmatrix} J_3 & 0 \\ * & \pm 2\delta s \end{pmatrix}. \quad (28)$$

Indeed the derivative of the last component in direction j is

$$2\beta(C(n \pm \delta j)k, C(j)k) = \pm 2\delta s$$

by (22) and (25). Thus the determinant of (28) is

$$\pm 24\rho^3(1+t^2)^6\delta s \neq 0. \quad (29)$$

The ordinary real implicit function theorem supplies smooth solutions $z_{\pm}(\varepsilon) \in E$, with $z_{\pm}(0) = 0$, on an interval about zero. These are actual local real paths, not merely formal expansions. Define points of the fixed whole space by

$$w_{\pm}(\varepsilon) = b_* + \varepsilon(n \pm \delta j + z_{\pm}(\varepsilon)). \quad (30)$$

Equations (15), (17), (20), and (27) imply the exact identities

$$Q(w_{\pm}(\varepsilon)) = \varepsilon R_- + \varepsilon^2 \delta^2 R_0. \quad (31)$$

For clarity, the first three components of (27) make the π -values on the two sides of (31) equal. The fourth component makes their ℓ -values equal, using (20) and $\ell(R_-) = 0$. Formula (19) then gives equality of every Gram entry. There is no omitted equation to impose.

Also, (22), (25), and smoothness give

$$D(w_{\pm}(\varepsilon)) = \pm \varepsilon \delta dD_{b_*}(j) + O(\varepsilon^2). \quad (32)$$

For every sufficiently small positive ε , their determinant signs are therefore respectively positive and negative.

The four labels in (4), evaluated on R_- , are $(0, 0, 16, 0)$, and on R_0 they are $(3/2, 1/2, 17/2, 1/2)$. Hence (31) gives

$$h(w_+(\varepsilon)) = h(w_-(\varepsilon)) = \frac{1}{2}\varepsilon^2\delta^2 > 0$$

for every positive ε for which the paths are defined. Thus, for sufficiently small positive ε ,

$$f(w_+(\varepsilon)) = \frac{1}{2}\varepsilon^2\delta^2 > 0, \quad f(w_-(\varepsilon)) = 0. \quad (33)$$

The normalized target Q/ε converges to the rank-one matrix R_- . The two small eigenvalues of the unnormalized matrix Q have scale $\varepsilon^2\delta^2$. No estimate uniform only near a fixed positive definite normalized target is used in (27)–(33).

A.6 Simultaneous agreement for every finite family of quadratic tests

Here is the exact finite-family statement:

For the fixed whole-space function (5), every finite family of real polynomials of degree at most two has two points of W on which all its members have identical signs, while f is positive at one point and zero at the other.

Fix an arbitrary such family $\mathcal{T}$. For a member $p = c + \lambda + q$, with homogeneous components of degrees zero, one and two, consider

$$p(\rho b(t)) = c + \rho \lambda(b(t)) + \rho^2 q(b(t)). \quad (34)$$

If this polynomial in (ρ, t) were identically zero, comparison of powers of ρ , followed by (13), would give $c = 0, \lambda = 0, q \in I_2$. Thus, for each member of $\mathcal{T} \setminus I_2$, at least one coefficient in t in (34) is a nonzero polynomial in ρ . Choose one $\rho > 0$ avoiding the finitely many roots of these finitely many coefficient polynomials. For this fixed ρ , every corresponding polynomial (34) is nonzero as a polynomial in t .

For each $q \in \mathcal{T} \cap (I_2 \setminus H_2)$, (14) says that $q(b'(t))$ is a nonzero polynomial in t . Choose a single real t avoiding its roots for all these members, and avoiding the roots of all the nonzero polynomials (34). Both choices can be made in any prescribed open intervals of positive ρ and real t . With $b_* = \rho b(t)$, we now have

$$\begin{aligned} p(b_*) &\neq 0 & (p \in \mathcal{T} \setminus I_2), \\ q(b') &\neq 0 & (q \in \mathcal{T} \cap (I_2 \setminus H_2)). \end{aligned} \tag{35}$$

For a homogeneous quadratic $q \in I_2$, twice differentiating $q(b(t)) = 0$ gives $dq_b(b'') = -2q(b')$. The definition of n in (16), with the scaling $b_* = \rho b$, therefore gives

$$dq_{b_*}(n) = \frac{q(b')}{d_*}. \tag{36}$$

For every member in the second line of (35), this is nonzero. Choose one $\delta > 0$ sufficiently small that

$$\delta |dq_{b_*}(j)| < \frac{1}{2} |dq_{b_*}(n)| \tag{37}$$

for each of those finitely many members. If the family of constraints is empty, any positive δ may be used. Fix this δ , and then construct the two actual paths in Appendix A.5.

For $q \in \mathcal{T} \cap (I_2 \setminus H_2)$, their expansions are

$$q(w_{\pm}(\varepsilon)) = \varepsilon(dq_{b_*}(n) \pm \delta dq_{b_*}(j)) + O(\varepsilon^2). \tag{38}$$

By (37), the two leading coefficients have the same nonzero sign. Hence the two values in (38) have the same nonzero sign for sufficiently small positive ε . For $p \in \mathcal{T} \setminus I_2$, the same conclusion follows from $p(b_*) \neq 0$ and continuity.

Finally, every $q \in \mathcal{T} \cap H_2$ is an actual constant linear combination of the three Gram entries in (14). Its values on the two paths are exactly equal by (31), including when that common value is zero. This handles all extra quadratic equations, not just one selected separator.

Take a single sufficiently small positive ε for all these finitely many conditions, for both implicit solutions, and for (32). All members of $\mathcal{T}$ then have identical signs at the two points, whereas (33) gives different signs of f . The quantifier order is:

$$\text{fixed } W, f; \quad \mathcal{T}; \quad \rho; \quad t; \quad \delta > 0; \quad \text{the two implicit solutions}; \quad \varepsilon > 0. \tag{39}$$

Neither the function nor its vector space depends on $\mathcal{T}$.

A.7 Excluding polynomial leaves of arbitrary degrees

The degree-two tests in Appendix A.6 do not impose a degree bound on the polynomials in a proposed lattice expression. The following limiting argument removes all such bounds.

Suppose a positively homogeneous degree-two function g on a real vector space has an expression

$$g = \mathcal{L}(p_1, \dots, p_m), \tag{40}$$

where $\mathcal{L}$ is a finite nonempty expression using minimum and maximum, and each p_i is any real polynomial. Write

$$p_i = c_i + \lambda_i + q_i + \text{terms of degrees at least three},$$

allowing zero components. At each fixed point w , the limit of $\tau^{-2}p_i(\tau w)$, as $\tau \rightarrow 0^+$, exists in the extended real line and is

$$L_i(w) = \begin{cases} \operatorname{sgn}(c_i) \infty, & c_i \neq 0, \\ \operatorname{sgn}(\lambda_i(w)) \infty, & c_i = 0, \lambda_i(w) \neq 0, \\ q_i(w), & c_i = 0, \lambda_i(w) = 0. \end{cases} \quad (41)$$

Positive scalar multiplication commutes with minimum and maximum. Thus homogeneity makes (40), after scaling, equal to $g(w) = \mathcal{L}(\tau^{-2}p_1(\tau w), \dots, \tau^{-2}p_m(\tau w))$ for every $\tau > 0$. Finite minimum and maximum are continuous on the extended real line; passing to the limits gives

$$g(w) = \mathcal{L}(L_1(w), \dots, L_m(w)). \quad (42)$$

The sign function from the extended real line to the ordered set $\{-1, 0, 1\}$ commutes with minimum and maximum. By (41)–(42), the sign of $g(w)$ is therefore determined by the signs of the finite family of linear and homogeneous quadratic polynomials

$$\{\lambda_i, q_i : 1 \leq i \leq m\}, \quad (43)$$

together with the fixed constants c_i . This conclusion also covers vanishing linear or quadratic components and infinite intermediate limits.

Apply the argument to the fixed function f , using (10). Appendix A.6 applied to (43) gives two points on which all those signs agree and the sign of f differs. This contradicts (42). Hence (5) has no polynomial lattice expression with arbitrary finite polynomial leaves.

Higher-degree test polynomials, including the determinant itself, may distinguish them; (41)–(43) explain why that fact cannot produce a lattice representation of this degree-two homogeneous function.